Question 18 — 1 mark

Given $f(x) = x - 1$, identify a point on the graph of $y = \sqrt{f(x)}$.

a) $(0,-1)$

b) $(3,2)$

c) $(1,0)$

d) $(0,1)$

Question 19 — 1 mark

Identify the total number of possible arrangements of 6 adults and 4 children seated in a row if the children must sit together.

a) $6!4!$

b) $7!4!$

c) $10!$

d) $6!$

Question 20 — 1 mark

Identify the exact value of $\sec\left(-\frac{7\pi}{3}\right)$.

a) -2

b) $-\frac{2}{\sqrt{3}}$

c) $\frac{2}{\sqrt{3}}$

d) 2

Question 21 1 mark

Given $\log_x\left(\frac{1}{25}\right) = -2,$ identify the value of x.

a) -5

b) $-\frac{1}{5}$

c) $\frac{1}{5}$

d) 5

Question 22 1 mark

Identify the equation for all of the asymptotes on the graph of $y = \tan x$.

a) $x = k\pi,\ k \in \mathbb{Z}$

b) $x = 2k\pi,\ k \in \mathbb{Z}$

c) $x = \frac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$

d) $x = \frac{\pi}{2} + 2k\pi,\ k \in \mathbb{Z}$

Question 23 1 mark

If $p(x) = 3(m)(x+1)^2$ is a cubic function with a y-intercept of $-12,$ identify the missing factor, m.

a) $m = x - 4$

b) $m = x + 4$

c) $m = x + 12$

d) $m = x - 12$

Question 24 — 1 mark

Identify the number of negative terms in the binomial expansion of $(x-y)^5$.

a) 2

b) 3

c) 5

d) 6

Question 25 — 1 mark

Given $f(x)=x^2$, identify which equation represents the graph of $y=f(x)$ after a translation of 5 units to the left.

a) $y=(x+5)^2$

b) $y=(x-5)^2$

c) $y=x^2-5$

d) $y=x^2+5$

Question 26 — 1 mark

When a polynomial, $p(x)$, is divided by $(x-7)$, the remainder is 24. Identify the only statement that must be true.

a) $x=7$ is a zero of $p(x)$

b) $p(7)=24$

c) $x=24$ is a zero of $p(x)$

d) the y-intercept is 24

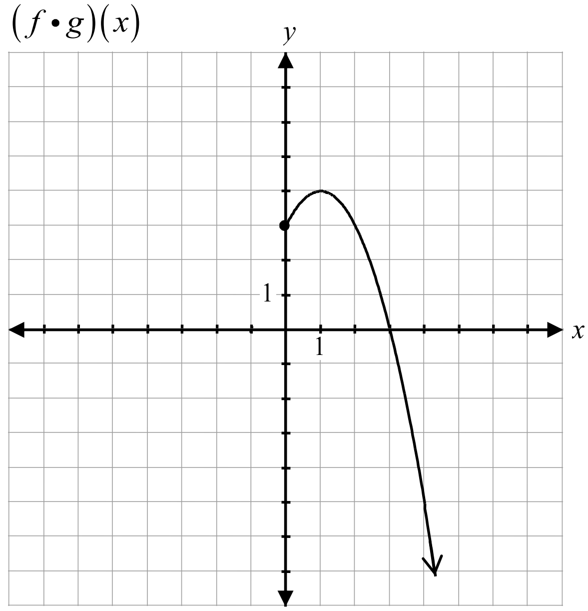
Question 27 1 mark 118

Given $f(x)=\dfrac{(2x+1)(x-8)}{(x-8)(x+4)}$, state the equation(s) of the vertical asymptote(s).

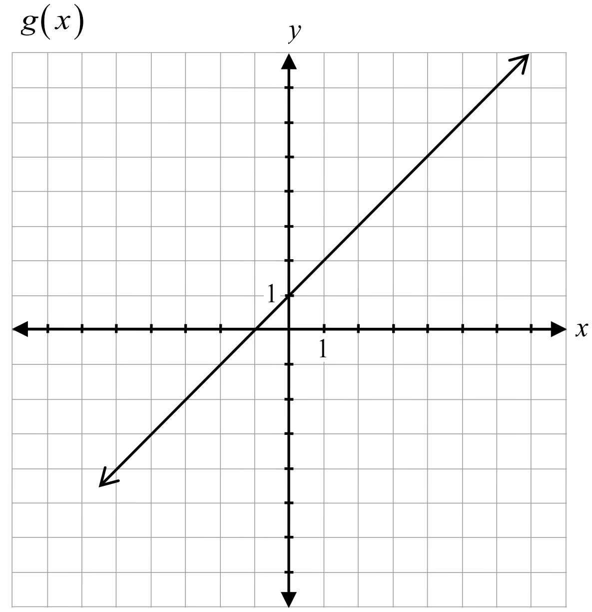
Question 28 2 marks 119

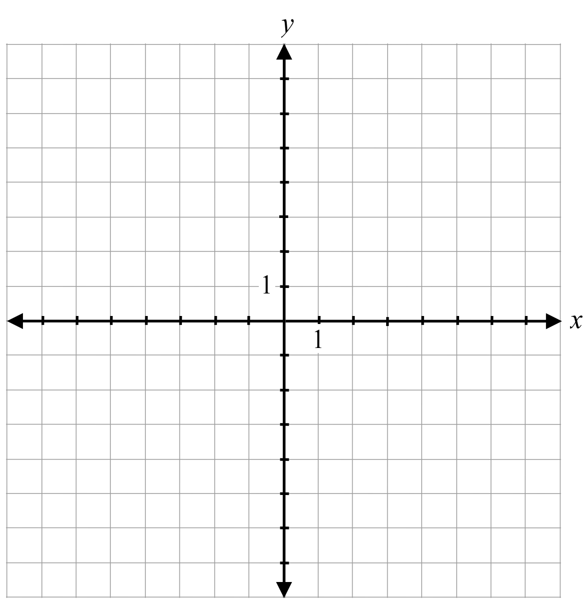
Given the graph of $(f \bullet g)(x)$ and $g(x)$, sketch the graph of $f(x)$.

$(f \bullet g)(x)$



$g(x)$





Question 29 1 mark 120

Brian was asked to state the zeros of the polynomial $p(x)=(x+2)(x-5)(x-1)$.

Brian's response:

Zeros : $(x+2),(x-5),(x-1)$

Explain why his response is incorrect.

Question 30 2 marks

121

Simplify ${}_{n+3}C_2$.

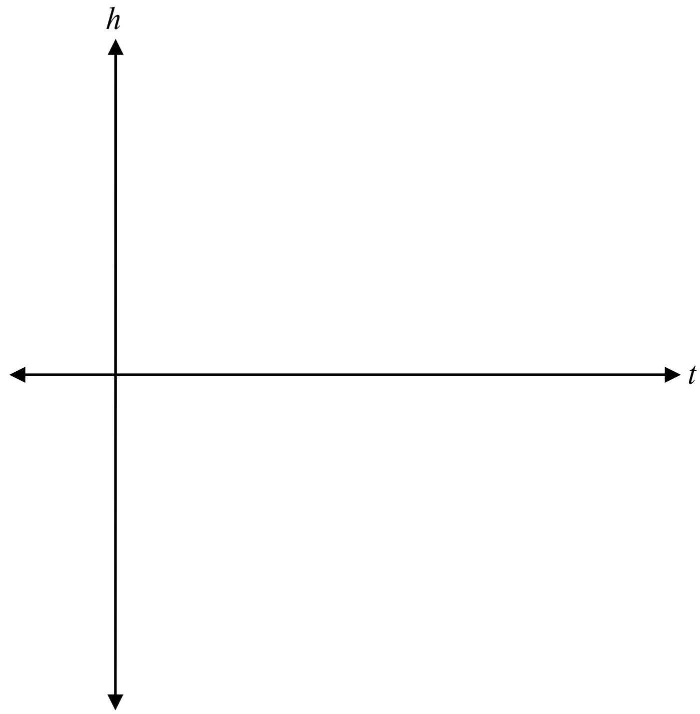
Question 31 2 marks 122

Verify that the equation $2\cos^2 x = \sin x + 1$ is true for $x = \frac{\pi}{6}$.

Question 32 a) 2 marks b) 1 mark 123 124

The height of a fish jumping out of the water can be modelled by the function $h(t) = -t(t-1)(t-4)(t-5)$ where $h(t)$ is the height of the fish above or below the water in cm, and t is the time in seconds, $t \geq 0$.

a) Sketch a graph representing the height of the fish with respect to time over the interval $[0, 5]$.



b) State, from the graph in a), the total number of seconds that the fish is above the water.

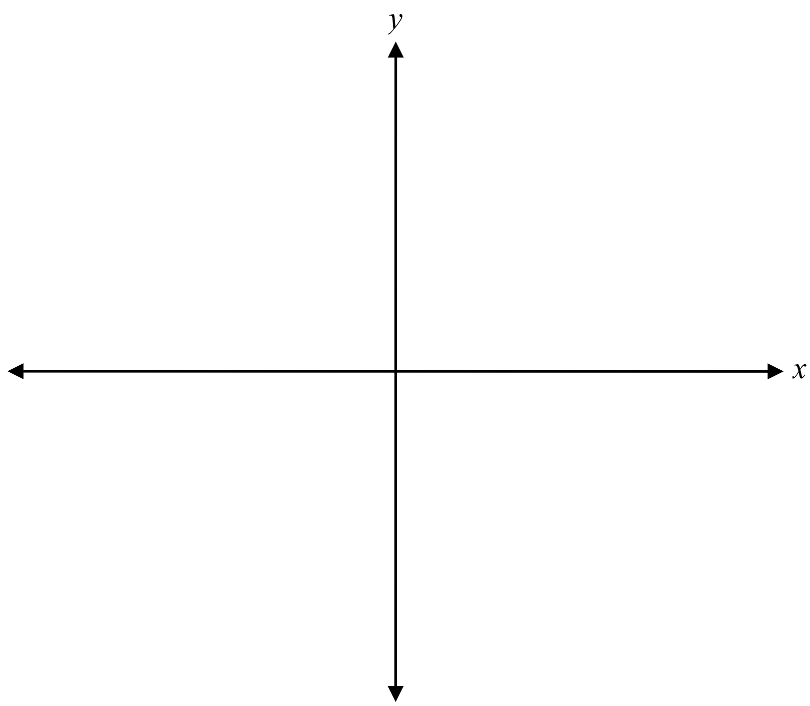
Question 33 1 mark

125

Describe the behaviour of the graph of $y = 5^x + 4$ as it approaches $y = 4$.

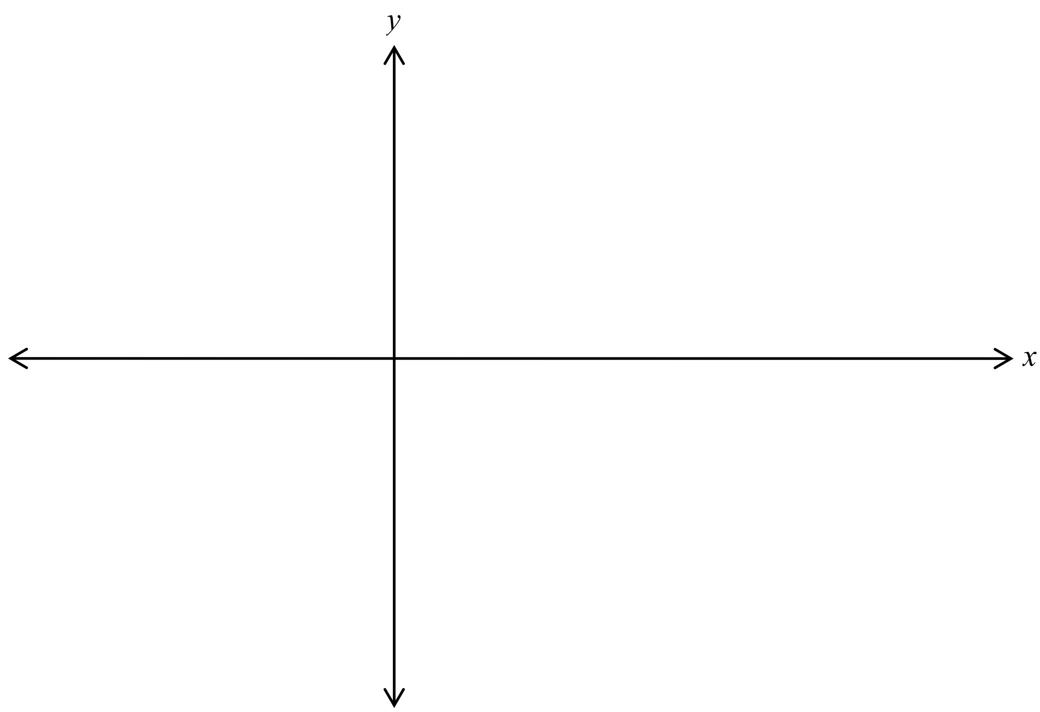
Question 34   1 mark   126

Sketch the angle of 6 radians in standard position.



Question 35 4 marks

Sketch the graph of the function $y = 4\sin\left(\frac{\pi}{3}x\right) - 2$ over the domain $[-3, 6]$.



Question 36

a) 1 mark b) 1 mark 128 129

Given $f(x) = 3x - 12$ and $g(x) = x - 4$,

a) determine the equation of $h(x) = \left(\dfrac{f}{g}\right)(x)$.

$h(x) =$ ______________________________

b) describe what the non-permissible value represents on the graph of $h(x)$.

Question 37 2 marks 130

Determine an equation of a radical function, $f(x)$, with a domain of $x \geq 5$ and a range of $y \geq -2$.

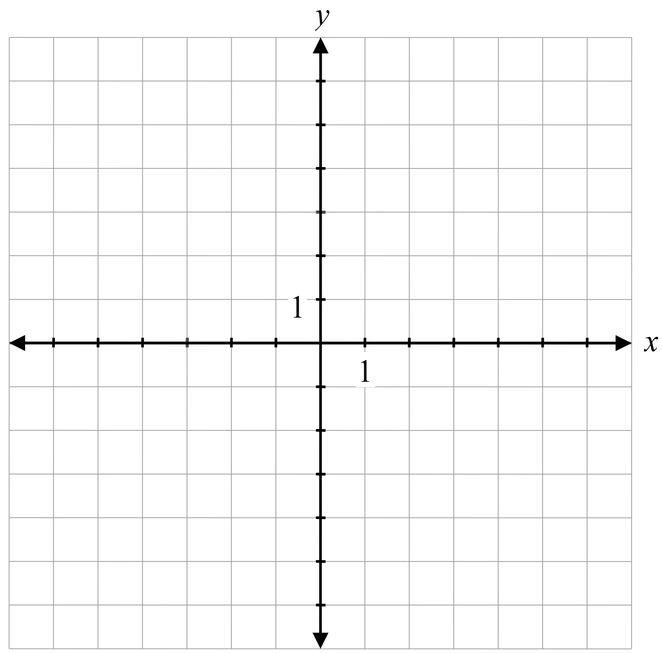
$f(x) =$ ______________________________

Question 38 3 marks 131

Determine the exact value of $\cos\left(\frac{17\pi}{12}\right)$.

Question 39 4 marks

Sketch the graph of $y = \frac{1}{2}\sqrt{-x} + 1$.



Question 40 2 marks

Given $f(x) = \frac{3x}{4} + 9,$ determine the equation of $f^{-1}(x)$.

Question 41 1 mark 134

Describe the end behaviour of the polynomial function $p(x) = -(x-2)(x+3)^2$.

Question 42 a) 2 marks b) 1 mark 135 136

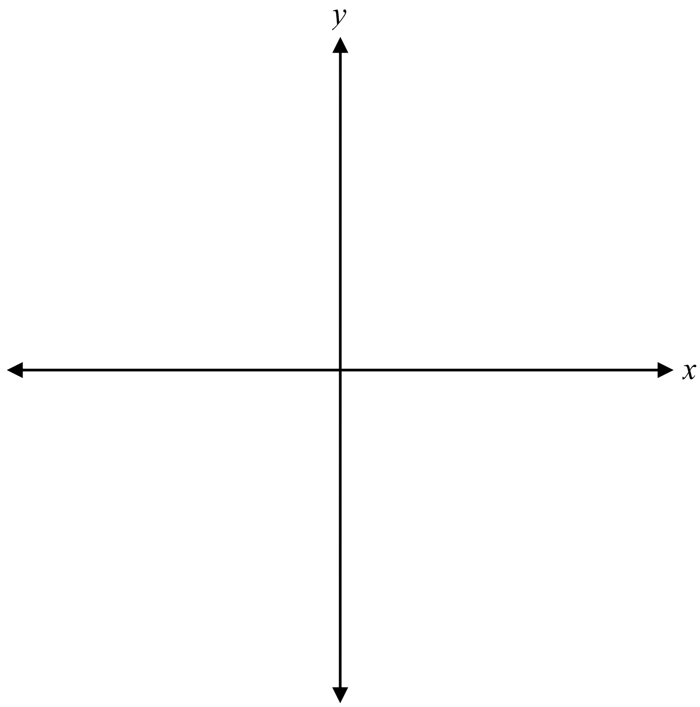
Given $\csc\theta = -4$ and θ is in quadrant IV,

a) determine the exact value of $\cos\theta$.

b) determine the exact value of $\cot\theta$.

Question 43 2 marks

Sketch the graph of the function $f(x) = \dfrac{10}{x^2 + 2}$.



Question 44 1 mark

Explain why only one of the following equations could be solved algebraically without using logarithms.

$3^{5x} = 6^{2x-1}$ or $16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$

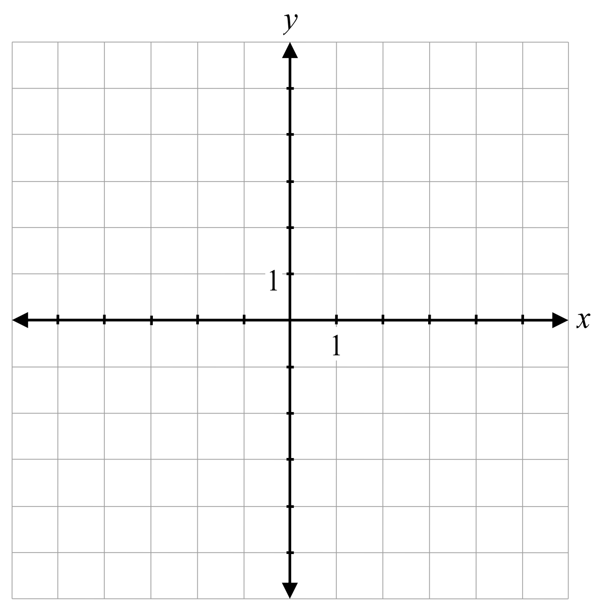
Question 45 1 mark 139

Given a graph of $y = f(x)$, describe how to sketch the graph of $y = |f(x)|$.

Question 46 2 marks

140

Sketch the graph of $y = \log_3(x+4)$.



Question 47 3 marks

Solve, algebraically.

$$\log x + \log 4 - \log(x-2) = \log 5$$

Question 48 2 marks

142

Given $\sin\theta = \frac{1}{2}$, determine all possible values of θ over the interval $[-2\pi, 2\pi]$.

No marks will be awarded for work done on this page.